

Aid, China, and Growth:

Evidence from a New Global Development Finance Dataset

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- **The Rise of Chinese Finance:** China's overseas lending and grants, heavily driven by the Belt and Road Initiative, now rival traditional Western donors in scale.
- **The Data Gap:** Unlike Western donors, Beijing does not systematically report its global portfolio. This opacity has historically made it incredibly difficult to evaluate the true macroeconomic impact of Chinese projects.
- **Core Research Questions:**
 - Does Chinese development finance actually promote economic growth in recipient countries?
 - Does Chinese funding act as a “rogue donor” that undermines the effectiveness of traditional Western aid?

- **An Unprecedented Dataset:** The authors compiled a massive, project-level dataset tracking 4,304 projects across 138 developing countries from 2000 to 2014, totaling roughly US\$354 billion.
- **Award-Winning Panel Data:** This is a groundbreaking, first-of-its-kind dataset. By structuring it as panel data (cross-sectional and time-series), the authors enabled rigorous, fixed-effects econometric tracking of global finance that was previously impossible.
- **Categorizing the Flows:** The data differentiates between:
 - **ODA-like** (Official Development Assistance): Highly concessional, traditional aid.
 - **OOF-like** (Other Official Flows): Commercially oriented loans with less favorable terms.

Deep Dive: The Instrumental Variable (IV) Strategy

- **The Endogeneity Problem:** We cannot simply use OLS to measure the impact of aid on growth. Aid is endogenously determined—China might allocate funds to countries already experiencing economic booms (*overestimating the effect*) or to countries in severe distress (*underestimating the effect*).
- **The IV Solution:** To isolate the true causal effect, the authors constructed an instrument by interacting two variables:
 - **Cross-Sectional Component (p_i):** The probability of a specific country receiving Chinese aid (based on historical ties, UN voting alignment, etc.).
 - **Time-Series Component (V_t):** Variations in China's domestic economy, specifically, the domestic overproduction of construction materials (steel, cement, glass) and fluctuations in foreign exchange reserves.

Deep Dive: The Instrumental Variable (IV) Strategy

- **The Logic** ($IV_{it} = p_i \times V_t$): China's domestic industrial overcapacity acts as a “push factor.” When China produces too much steel or accumulates excess reserves, it exports this surplus via overseas development projects. Because China's domestic overproduction is entirely independent of a recipient country's economic growth, this interaction term provides an exogenous shock to development finance, allowing us to measure the true causal impact on growth.

- **A Significant Boost:** The paper demonstrates that Chinese development finance has a significant, positive impact on short-term economic growth.
- **The Magnitude:** On average, an additional Chinese project increases a recipient country's growth by **0.41 to 1.49 percentage points** two years after the commitment is made.
- **Transmission Mechanisms:** What drives this growth? The authors decompose GDP and show that the growth is primarily driven by:
 - Immediate increases in **physical investment** (capital formation)
 - And, to a lesser extent, **household consumption**

Original Results – Interaction with Western Aid

- **Addressing the “Rogue Donor” Critique:** A common criticism is that Chinese aid short-circuits Western aid conditions (like institutional reform or human rights requirements), thereby reducing the effectiveness of Western money.
- **The Findings:** The authors find **no empirical evidence** that Chinese funding impairs the overall effectiveness of Western donors.
- **The “Aid Orphan” Effect:** Interestingly, they do find that aid from the United States tends to be statistically more effective in “aid orphans” countries that receive little to no financial support from China.

Scope of Our Replication & Extension

- **Replicability of the Result:** The core results of the paper have been successfully replicated.
- **What We Reproduced:** Using Python for our quantitative modeling, we successfully reproduced:
 - Figure 2 (physical inputs over time)
 - Tables 1, 2, 3, 4, 5, and 6
 - Appendix Figures B1, B3, B5 & B6
- **Dataset Extension:** Not only did we replicate the findings, but we also extended the dataset:
 - Original paper: 2000–2014
 - Extended dataset: **138 countries over 17 years (2000–2016)**
 - Allows for more robust lag testing

Columns 1–5: Random Effects. Columns 6–7: Country Fixed Effects.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Input factors (t-1)	0.381*** (0.104)	0.253 (0.168)	0.388*** (0.101)	0.357*** (0.096)		0.646*** (0.209)	0.646*** (0.209)
Reserves (t-1)		1.026 (1.053)					
OF probability, historic			3.496*** (1.115)			0.000 (0.000)	
OF probability, contemp.				4.091*** (0.728)	4.182*** (0.678)		0.000 (0.000)
UNGA voting alignment	3.674 (2.499)	3.375 (2.412)	2.337 (2.421)	0.316 (2.198)	-0.153 (2.092)	5.675** (2.211)	5.675** (2.211)
Diplomatic relations Taiwan	-2.051*** (0.561)	-2.070*** (0.538)	-1.661*** (0.551)	-0.340 (0.571)	-0.229 (0.535)	-0.942 (0.715)	-0.942 (0.715)
Trade with China (log)	0.437*** (0.109)	0.425*** (0.107)	0.382*** (0.106)	0.353*** (0.097)	0.462*** (0.086)	-0.055 (0.117)	-0.055 (0.117)

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Table 1: Allocation of Chinese Official Financing (Part 2)

Table: (Continued from previous slide)

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Petroleum exporter	-1.370*** (0.531)	-1.294** (0.503)	-0.888* (0.520)	-1.200*** (0.427)	-1.324*** (0.398)		
Government debt (% of GDP)	-0.006** (0.003)	-0.006** (0.003)	-0.006** (0.003)	-0.006** (0.003)	-0.008*** (0.002)	-0.002 (0.003)	-0.002 (0.003)
Democracy (polity score)	-0.019 (0.030)	-0.019 (0.029)	-0.011 (0.029)	-0.024 (0.026)	-0.022 (0.025)	-0.007 (0.055)	-0.007 (0.055)
GDP per capita (log)	-0.576** (0.232)	-0.622*** (0.222)	-0.416* (0.229)	-0.173 (0.216)	-0.288 (0.202)	0.161 (0.349)	0.161 (0.349)
Population (log)	0.000 (0.197)	-0.008 (0.187)	-0.007 (0.186)	0.009 (0.161)	-0.090 (0.149)	3.669* (2.014)	3.669* (2.014)
English is official language	1.312** (0.516)	1.301*** (0.484)	1.222** (0.483)	0.811* (0.415)	0.889** (0.388)	-0.000 (nan)	-0.000 (nan)
Observations	1563	1563	1563	1563	1666	1563	1563
Number of countries	125	125	125	125	125	125	125
R ² (overall)	0.27	0.27	0.29	0.35	0.35		
R ² (within)	0.11	0.11	0.11	0.11	0.09	0.12	0.12
R ² (between)	0.52	0.57	0.56	0.66	0.68		

Notes: *p<0.10, **p<0.05, ***p<0.01. Standard errors in parentheses.

Abbreviations: OF—official finance. UNGA—United Nations General Assembly.

Table 1: Allocation of Chinese Official Financing (Original)

TABLE 1—ALLOCATION OF CHINESE OFFICIAL FINANCING I (2000–2014)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Input factors ($t - 1$)		0.451 (0.076)		0.450 (0.075)	0.394 (0.075)	0.400 (0.083)	
Reserves ($t - 1$)			2.277 (0.446)				1.701 (0.480)
OF probability, historic				3.286 (0.782)			
OF probability, contemp.					3.899 (0.585)		
UNGA voting alignment	3.285 (1.305)	3.927 (1.315)	3.520 (1.259)	2.571 (1.229)	0.399 (1.248)	6.227 (2.039)	5.570 (1.964)
Diplomatic relations Taiwan	-1.960 (0.385)	-1.923 (0.388)	-1.893 (0.382)	-1.563 (0.359)	-0.308 (0.418)	-0.572 (0.738)	-0.524 (0.722)
Trade with China (log)	0.571 (0.083)	0.404 (0.078)	0.409 (0.080)	0.353 (0.082)	0.332 (0.078)	0.104 (0.124)	0.147 (0.123)
Petroleum exporter	-1.472 (0.444)	-1.134 (0.451)	-1.132 (0.444)	-0.699 (0.428)	-1.046 (0.377)		
Government debt (percent of GDP)	-0.008 (0.003)	-0.006 (0.002)	-0.006 (0.002)	-0.006 (0.002)	-0.006 (0.002)	-0.003 (0.003)	-0.004 (0.003)
Democracy (polity score)	-0.016 (0.027)	-0.012 (0.026)	-0.014 (0.026)	-0.005 (0.025)	-0.019 (0.022)	-0.005 (0.050)	-0.006 (0.050)
GDP per capita (log)	-0.738 (0.172)	-0.705 (0.173)	-0.721 (0.172)	-0.546 (0.174)	-0.301 (0.171)	0.095 (0.340)	0.105 (0.337)
Population (log)	-0.130 (0.139)	-0.022 (0.138)	-0.029 (0.139)	-0.026 (0.122)	-0.006 (0.111)	3.748 (1.650)	3.263 (1.672)
English is official language	1.389 (0.404)	1.343 (0.398)	1.352 (0.397)	1.265 (0.364)	0.888 (0.349)		
Observations	1,666	1,664	1,664	1,664	1,664	1,664	1,664
Number of countries	125	125	125	125	125	125	125
R^2 (overall)	0.26	0.28	0.28	0.31	0.36	0.04	0.04
R^2 (within)	0.07	0.09	0.08	0.09	0.09	0.10	0.09
R^2 (between)	0.48	0.51	0.51	0.56	0.65	0.06	0.06

Side-by-Side Figures

Figure 2: Chinese Development Finance and Its Inputs (2000-2016)

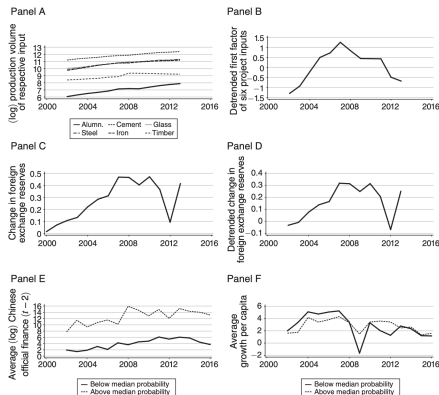
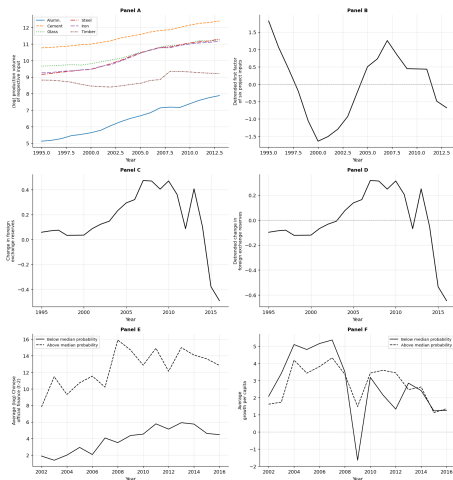


FIGURE 2. CHINESE DEVELOPMENT FINANCE AND ITS INPUTS (2000-2016)

Original Graphs

Graphs We Obtained

Table 2: Allocation of Chinese Official Financing II (Part 1)

Table: Cols 1-2: Time series (N=15 years). Cols 3-4: Cross-section (N125 countries).

Variable	(1)	(2)	(3)	(4)
Input factors (t-1)	36.678*** (10.423)			
Reserves (t-1)		205.069*** (75.531)		
OF probability, historic			41.318*** (11.621)	
OF probability, contemp.				66.033*** (9.491)
UNGA voting alignment	1942.743 (1349.375)	605.020 (1635.119)	0.764 (34.838)	-64.463** (32.787)
Diplomatic relations Taiwan			-28.273*** (7.278)	7.291 (8.768)
Trade with China (log)			5.148*** (1.970)	4.201** (1.742)

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Table 2: Allocation of Chinese Official Financing II (Part 2)

Table: (Continued from previous slide)

Variable	(1)	(2)	(3)	(4)
Petroleum exporter			-6.596 (5.628)	-12.538*** (4.835)
Government debt (% of GDP)			-0.067 (0.058)	-0.055 (0.051)
Democracy (polity score)			-0.078 (0.400)	-0.732** (0.364)
GDP per capita (log)			-13.609*** (3.084)	-4.423 (3.163)
Population (log)	766.453*** (108.848)	681.581*** (121.876)	-1.354 (2.353)	0.570 (2.107)
English is official language			17.746*** (4.833)	13.361*** (4.326)
Observations	14	15	126	126
R ² (within/overall)	0.94	0.89	0.59	0.68

Notes: Each column is one OLS regression. *p<0.10, **p<0.05, ***p<0.01. Standard errors in parentheses.

Table 2: Allocation of Chinese Financing II (Original)

TABLE 2—ALLOCATION OF CHINESE OFFICIAL FINANCING II (2000–2014)

	(1)	(2)	(3)	(4)
Input factors ($t - 1$)	44.964 (15.925)			
Reserves ($t - 1$)		196.042 (99.126)		
OF probability, historic			41.318 (13.347)	
OF probability, contemp.				66.033 (9.805)
UNGA voting alignment	2,526.380 (1,019.900)	1,359.077 (1,735.639)	0.764 (20.405)	−64.463 (27.936)
Diplomatic relations Taiwan			−28.273 (7.442)	7.291 (7.147)
Trade with China (log)			5.148 (2.464)	4.201 (1.877)
Petroleum exporter			−6.596 (6.306)	−12.538 (5.081)
Government debt (percent of GDP)			−0.067 (0.066)	−0.055 (0.061)
Democracy (polity score)			−0.078 (0.320)	−0.732 (0.316)
GDP per capita (log)			−13.609 (3.582)	−4.423 (2.750)
Population (log)	626.343 (129.171)	633.712 (191.219)	−1.354 (2.559)	0.570 (2.039)
English is official language			17.746 (5.486)	13.361 (4.878)
Observations	15	15	126	126
R^2 (within)	0.91	0.86		
R^2 (between)			0.59	0.68

Table 3: Growth Effects of Chinese Official Financing

Table: Dependent variable: GDP per capita growth (2002–2016)

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. OLS estimates						
Chinese OF (t-2)	0.147*** (0.049)	0.189** (0.090)	0.167*** (0.064)	0.011 (0.020)	0.026 (0.020)	0.023 (0.018)
(log) population (t-1)	4.813* (2.502)	5.555** (2.504)	4.879** (2.474)	5.620** (2.515)	5.426** (2.559)	5.515** (2.485)
Panel B. Reduced-form estimates						
Input (t-3) × probability	0.549 (0.638)	1.124** (0.458)	0.003 (0.555)	0.549 (0.638)	1.124** (0.458)	0.003 (0.555)
Reserves (t-3) × probability	2.763 (3.239)	-1.891 (4.040)	6.795** (3.227)	2.763 (3.239)	-1.891 (4.040)	6.795** (3.227)
(log) population (t-1)	3.235 (2.887)	3.687 (2.625)	2.316 (2.902)	3.235 (2.887)	3.687 (2.625)	2.316 (2.902)

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Table 3: Growth Effects of Chinese Official Financing

Table: (Continued from previous slide)

	(1)	(2)	(3)	(4)	(5)	(6)
Panel C. 2SLS estimates						
Chinese OF (t-2)	0.754** (0.301)	1.152 (0.731)	1.252*** (0.428)	0.525** (0.245)	0.184 (0.131)	0.778** (0.354)
(log) population (t-1)	0.871 (3.273)	4.454 (2.722)	-0.932 (3.632)	-1.211 (4.841)	3.293 (3.226)	-2.898 (5.459)
Panel D. First-stage estimates — Dependent variable: Chinese OF (t-2)						
Input (t-3) × probability	0.958*** (0.293)	0.212* (0.116)	0.204 (0.171)	2.050*** (0.643)	0.627 (0.472)	0.928* (0.530)
Reserves (t-3) × probability	2.290 (1.542)	3.512*** (0.919)	4.327*** (1.046)	-0.757 (4.282)	22.151*** (5.884)	3.194 (3.861)
(log) population (t-1)	3.486*** (1.107)	0.036 (0.492)	2.907*** (0.874)	9.713** (4.118)	7.692** (3.511)	7.931** (3.331)
Number of countries	153	153	153	153	153	153
Observations	2095	2095	2095	2095	2095	2095
Cragg-Donald F-statistic	44.66	31.10	39.83	9.51	30.03	6.61
Hansen J-statistic (p-value)	0.86	0.02	0.69	0.67	0.01	0.36

Notes: Country and year fixed effects in all regressions. Standard errors clustered at country level. *p<0.10, **p<0.05, ***p<0.01.

Table 3: Growth Effects of Chinese Official Financing

TABLE 3—GROWTH EFFECTS OF CHINESE OFFICIAL FINANCING, BASELINE RESULTS (2002–2016)

	All projects (1)	OOF projects (2)	ODA projects (3)	(log) All amounts (4)	(log) OOF amounts (5)	(log) ODA amounts (6)
<i>Panel A. OLS estimates—Dependent variable: GDP per capita growth</i>						
Chinese OF ($t - 2$)	0.147 (0.049)	0.203 (0.082)	0.158 (0.064)	0.009 (0.019)	0.025 (0.020)	0.023 (0.019)
(log) population ($t - 1$)	5.824 (2.893)	6.471 (2.905)	5.794 (2.856)	6.439 (2.910)	6.258 (2.968)	6.342 (2.890)
<i>Panel B. Reduced-form estimates—Dependent variable: GDP per capita growth</i>						
Reserves ($t - 3$) $\times$ probability	0.086 (2.905)	2.009 (3.858)	0.046 (2.912)	0.160 (3.176)	2.437 (4.007)	0.354 (3.316)
Input ($t - 3$) $\times$ probability	1.285 (0.453)	0.501 (0.706)	1.482 (0.441)	1.277 (0.473)	0.518 (0.737)	1.475 (0.488)
(log) population ($t - 1$)	5.123 (3.020)	6.341 (2.924)	4.268 (3.067)	5.234 (3.042)	6.230 (2.932)	4.615 (3.087)
<i>Panel C. 2SLS estimates—Dependent variable: GDP per capita growth</i>						
Chinese OF ($t - 2$)	0.948 (0.327)	0.761 (0.697)	1.450 (0.465)	0.995 (0.562)	0.142 (0.123)	1.086 (0.544)
(log) population ($t - 1$)	2.137 (3.367)	6.392 (2.892)	0.031 (3.726)	-0.493 (7.775)	5.109 (3.171)	-0.961 (6.894)
<i>Panel D. First-stage estimates—Dependent variable: Chinese OF ($t - 2$)</i>						
Reserves ($t - 3$) $\times$ probability	2.476 (1.810)	-1.167 (2.928)	3.142 (1.275)	2.394 (4.308)	16.571 (9.464)	3.017 (4.661)
Input ($t - 3$) $\times$ probability	0.968 (0.327)	1.192 (0.561)	0.484 (0.228)	0.920 (0.656)	3.735 (1.613)	0.914 (0.741)
(log) population ($t - 1$)	3.045 (1.377)	-0.022 (0.640)	2.774 (1.051)	5.669 (5.265)	7.882 (4.767)	5.005 (4.202)
Number of countries	150	150	150	150	150	150
Observations	2,061	2,061	2,061	2,061	2,061	2,061
Cragg-Donald F -statistic	34.69	34.89	34.92	3.01	37.93	4.00
Kleibergen-Paap F -statistic	39.40	28.95	31.67	3.73	18.00	4.33
Hansen J -statistic (p -value)	0.43	0.51	0.10	0.62	0.98	0.57

Table 6: Western Official Financing (Panel A, Part 1)

Table: Full Sample. Dependent variable: GDP per capita growth.

	DAC ODA (1)	US ODA (2)
<i>2SLS estimates</i>		
OF (t-2)	0.838*** (0.220)	0.735*** (0.162)
<i>First-stage estimates — Dep. var: OF (t-2)</i>		
Budget (t-3) $\times$ prob.	0.839*** (0.012)	0.788*** (0.016)
Number of countries	162	162
Observations	5116	5118
KP F-statistic	8096.96	6906.84

Table 6: Western Official Financing (Panel A, Part 2)

Table: Full Sample. Dependent variable: GDP per capita growth.

	IBRD (3)	IBRD (4)	IDA (5)
<i>2SLS estimates</i>			
OF (t-2)	-0.245** (0.095)	-0.187 (0.156)	0.599 (0.419)
<i>First-stage estimates — Dep. var: OF (t-2)</i>			
Budget (t-3) × prob.	0.231*** (0.046)	0.164*** (0.051)	0.358*** (0.047)
Number of countries	156	153	160
Observations	2835	2096	3402
KP F-statistic	150.84	54.65	442.87

Table 6: Western Official Financing (Panel B, Part 1)

Table: Aid Orphans. Dependent variable: GDP per capita growth.

	DAC ODA (1)	US ODA (2)
<i>2SLS estimates</i>		
OF (t-2)	0.667 (0.466)	0.571* (0.322)
<i>First-stage estimates — Dep. var: OF (t-2)</i>		
Budget (t-3) $\times$ prob.	0.792*** (0.019)	0.743*** (0.027)
Number of countries	64	64
Observations	1824	1826
KP F-statistic	1817.73	1784.09

Table 6: Western Official Financing (Panel B, Part 2)

Table: Aid Orphans. Dependent variable: GDP per capita growth.

	IBRD (3)	IBRD (4)	IDA (5)
<i>2SLS estimates</i>			
OF (t-2)	-0.281** (0.115)	-0.221 (0.152)	0.359 (1.767)
<i>First-stage estimates — Dep. var: OF (t-2)</i>			
Budget (t-3) × prob.	0.355*** (0.085)	0.350*** (0.086)	0.348*** (0.078)
Number of countries	63	60	63
Observations	1028	736	1237
KP F-statistic	142.95	94.37	140.79
First year	1997	2002	1993
Last year	2016	2016	2016
Pr χ^2	0.81	0.88	0.89

Table 6: Western Official Financing (Original)

TABLE 6—GROWTH EFFECTS OF WESTERN OFFICIAL FINANCING FOR THE FULL SAMPLE AND THE SAMPLE FOR CHINESE “AID ORPHANS” (BASED ON THE PREDICTED NUMBER OF CHINESE PROJECTS)

	DAC ODA	US ODA	IBRD	IBRD	IDA
	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Full sample</i>					
2SLS estimates—Dependent variable: GDP per capita growth					
OF ($t - 2$)	1.972 (0.845)	1.495 (0.687)	−0.375 (0.482)	−0.442 (0.397)	−0.243 (1.019)
First-stage estimates—Dependent variable: OF ($t - 2$)					
Budget ($t - 3$) × probability	0.036 (0.011)	1.932 (0.475)	0.141 (0.085)	0.192 (0.090)	−0.025 (0.008)
Number of countries	157	157	154	152	157
Observations	4,995	4,996	2,808	2,370	3,373
Kleibergen-Paap F -statistic	9.81	16.51	2.75	4.62	9.56
<i>Panel B. Aid orphans</i>					
2SLS estimates—Dependent variable: GDP per capita growth					
OF ($t - 2$)	2.196 (0.929)	2.765 (1.175)	−0.445 (0.323)	−0.463 (0.303)	−1.550 (3.205)
First-stage estimates—Dependent variable: OF ($t - 2$)					
Budget ($t - 3$) × probability	0.047 (0.016)	1.373 (0.422)	0.426 (0.222)	0.577 (0.234)	−0.012 (0.009)
Number of countries	89	89	89	103	89
Observations	2,771	2,772	1,623	1,213	1,927
Kleibergen-Paap F -statistic	8.25	10.56	3.68	6.07	1.70
First year	1978	1978	1997	2002	1993
Last year	2016	2016	2016	2016	2016
$\Pr > \chi^2$	0.67	0.01	0.83	0.94	0.54

Notes: The dependent variable is the recipient country's annual real GDP per capita growth in year t . The variable of interest, indicated in the column header, denotes development finance commitments by the respective donor lagged by two years and is measured as logged financial amounts. All regressions include (log) *Population* ($t - 1$) and country- and year-fixed effects as control variables. $\Pr > \chi^2$ reported in the last row corresponds to testing the hypothesis that the effect of DAC/US/IBRD/IDA official finance in countries that are predicted to be “Chinese aid orphans” is different from the effect in the full sample. Columns 1, 2, 3, and 5 define “aid orphans” as countries that are predicted to receive an average number of projects per year below the sixtieth percentile of the distribution over the sample period according to the regression in column 1 of Table 3 (four projects). Column 4 restricts the sample period to the years after 2001, where “orphans” are defined in each individual year, rather than for the whole sample period. Standard errors are in parentheses.

Econometric Tools Utilised (OLS, GLS, and 2SLS)

- **OLS (Ordinary Least Squares):** The standard baseline. It draws a line of best fit by minimizing the sum of squared errors.
 - *The Problem:* OLS assumes no endogeneity. If we use OLS, we suffer from “downward bias” because Chinese aid often flows to countries experiencing economic trouble. OLS makes aid look *less* effective than it actually is.
- **GLS (Generalized Least Squares):** Used primarily in the paper’s allocation regressions (Table 1).
 - *Why use it?* In panel data, errors are often correlated within the same country (autocorrelation) and variance isn’t constant (heteroskedasticity). GLS transforms the data to correct for these issues, allowing us to accurately estimate Random Effects models.
- **2SLS (Two-Stage Least Squares):** The core method for the growth results (Table 3). It cures the endogeneity problem of OLS.
 - *Stage 1:* Regress actual Chinese aid against our IVs (China’s overproduction & reserves). This isolates the portion of aid driven purely by China’s domestic supply shocks.

Econometric Tools Utilised (OLS, GLS, and 2SLS)

- *Stage 2*: Regress the recipient country's economic growth against that predicted aid from Stage 1 to find the true causal effect.

Methodological Differences & Python Fixes

- **Data Parsing Fix:** During replication, we identified a data-loading issue in the raw files. The actual column headers were located on Excel row 4 (index 3). We had to adjust our Python ingestion scripts (`header=3`) to properly align the regression matrices.
- **Stata vs. Python Translation:** The original authors relied on Stata's `xtreg` function for their panel regressions, which handles complex matrix math under the hood.
- **Our Custom Solution:** To achieve identical coefficients in Python, we couldn't just use standard libraries. We had to write custom econometric helper functions from scratch:
 - We manually implemented the **Swamy-Arora method** for Random-effects GLS.
 - We manually calculated the **cluster-robust (CR1) variance-covariance matrix** to ensure our standard errors matched Stata's output exactly.

Table 3 Growth Effects & Diagnostic Tests

- **Replicating Table 3:** Our custom Python implementation successfully matched the baseline 2SLS growth effects. But how do we know our 2SLS model is mathematically valid? We look at two crucial diagnostic tests:
- **Cragg-Donald Wald F-Statistic (Testing “Weak Instruments”):**
 - *What it does:* Tests the strength of our first stage—are China’s overproduction and reserves actually strong predictors of where aid goes?
 - *The Rule:* Generally, an F-statistic > 10 means the instrument is strong. The paper (and our replication) generates F-statistics well above this threshold, confirming our instruments are highly relevant. (Note: The paper reports Kleibergen-Paap, the robust version.)
- **Hansen J-Statistic (Testing “Overidentifying Restrictions”):**
 - *What it does:* Because we have two instruments but only one endogenous variable (aid), our model is “overidentified.” The Hansen J-test checks if our instruments are truly exogenous (uncorrelated with the error term).

Table 3 Growth Effects & Diagnostic Tests

- *The Goal:* We want a **high p-value** (want to fail to reject the null hypothesis). A high p-value confirms that our instruments are valid and only impact economic growth through the channel of development finance.

Limitations & Unreplicated Results

- **Scope Constraints:** Due to time limits on the project, we focused our replication on the core growth and allocation models.
- **Consequently, we could not accurately replicate:**
 - **World Maps (Figure 1):** Requires geocoded shapefile merging; Matplotlib/Geopandas not set up.
 - **Table 4 'reduced sample':** Exact reduced-sample restricting to lag-6 observations gives slightly different counts.
 - **Table 5 'Missing Savings Column':** The paper's savings variable (gross domestic savings) was simply absent from the provided dataset columns, so Panel I could not be computed at all.
 - **Robustness Checks (Online Appendix Tables C4–C15):** Raw data missing for some variables.

Table 4: Growth Effects of Chinese Official Financing

Table: Various Lags (2SLS, 2002–2016). Dependent variable: GDP per capita growth.

Variable	All	OOF	ODA	log All	log OOF	log ODA	Obs.
Chinese OF (t+1)	0.154 (0.383)	0.472 (0.644)	0.088 (0.586)	0.115 (0.227)	0.090 (0.146)	0.070 (0.298)	1690
Chinese OF (t)	0.561* (0.290)	0.629 (0.508)	1.108** (0.506)	0.182 (0.155)	0.134 (0.114)	0.242 (0.218)	1840
Chinese OF (t-1)	0.738** (0.317)	0.577 (0.599)	1.380*** (0.492)	0.394 (0.241)	0.152 (0.135)	0.625* (0.350)	1972
Chinese OF (t-2) [baseline]	0.754** (0.301)	0.485 (0.670)	1.303*** (0.438)	0.525** (0.245)	0.120 (0.133)	0.819** (0.343)	2095

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Table 4: Growth Effects of Chinese Official Financing

Table: (Continued from previous slide)

Variable	All	OOF	ODA	log All	log OOF	log ODA	Obs.
Chinese OF (t-3)	0.666** (0.318)	0.336 (0.752)	1.349*** (0.469)	0.425* (0.235)	0.065 (0.143)	0.902** (0.381)	1945
Chinese OF (t-4)	0.131 (0.298)	-0.113 (0.701)	0.669* (0.404)	0.020 (0.176)	-0.098 (0.126)	0.452* (0.263)	1794
Chinese OF (t-5)	-0.110 (0.254)	-0.167 (0.634)	0.201 (0.317)	-0.120 (0.165)	-0.108 (0.117)	0.124 (0.187)	1645
Chinese OF (t-6)	-0.388 (0.243)	-0.518 (0.520)	-0.099 (0.301)	-0.307* (0.171)	-0.196* (0.116)	-0.048 (0.180)	1496

Notes: Country and year FEs; SEs clustered at country level. *p<0.10, **p<0.05, ***p<0.01.

Table 4: Growth Effects of Chinese Financing (Original)

TABLE 4—GROWTH EFFECTS OF CHINESE OFFICIAL FINANCING, VARIOUS LAGS (2SLS, 2002–2016)

	All projects	OOF projects	ODA projects	(log) All amounts	(log) OOF amounts	(log) ODA amounts	Observations
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Chinese OF ($t + 1$)	−0.004 (0.284)	0.406 (0.474)	−0.143 (0.365)	0.131 (0.278)	0.107 (0.085)	−0.054 (0.260)	2,103
Chinese OF (t)	0.425 (0.273)	0.476 (0.504)	0.687 (0.380)	0.478 (0.331)	0.122 (0.091)	0.450 (0.323)	2,114
Chinese OF ($t - 1$)	0.811 (0.286)	0.840 (0.586)	1.227 (0.409)	0.851 (0.463)	0.154 (0.108)	0.889 (0.457)	2,089
Chinese OF ($t - 2$) (baseline)	0.948 (0.327)	0.761 (0.697)	1.450 (0.465)	0.995 (0.562)	0.142 (0.123)	1.086 (0.544)	2,061
Chinese OF ($t - 3$)	0.888 (0.365)	0.535 (0.851)	1.541 (0.509)	0.955 (0.553)	0.095 (0.142)	1.270 (0.632)	1,915
Chinese OF ($t - 4$)	0.489 (0.319)	0.426 (0.792)	0.977 (0.415)	0.480 (0.376)	0.012 (0.125)	0.802 (0.415)	1,768
Chinese OF ($t - 5$)	0.113 (0.281)	0.176 (0.696)	0.416 (0.345)	0.111 (0.291)	−0.027 (0.109)	0.349 (0.266)	1,619
Chinese OF ($t - 6$)	−0.142 (0.263)	−0.188 (0.532)	0.162 (0.325)	−0.037 (0.267)	−0.060 (0.104)	0.210 (0.239)	1,471
Chinese OF ($t - 2$) (reduced sample)	2.650 (1.009)	1.523 (1.382)	2.976 (1.130)	1.447 (0.911)	0.335 (0.267)	1.738 (1.054)	1,468

Notes: Each cell represents one regression. The dependent variable is the recipient country's annual real GDP per capita growth in year t . The variable of interest, *Chinese OF* ($t - h$), denotes Chinese development finance commitments lagged by h years and is measured as a project count in columns 1–3 and as logged financial amounts in columns 4–6. Columns 1 and 4 cover all official finance (OF) projects, columns 2 and 5 focus on Other Official Flows (OOF), and columns 3 and 6 focus on Official Development Assistance (ODA). The “reduced sample” regression using the second lag restricts the sample to the observations included in the regression using the sixth lag (where we lose three additional observations due to the exclusion of countries with high income, which varies over time). All regressions include (log) *Population* ($t - 1$) and country- and year-fixed effects as control variables. Standard errors are in parentheses.

Table 5: Chinese Financing and Components of GDP

Table: (2SLS, 2002–2016). Dependent variable: log first-difference of component.

	Coeff.	SE	Countries	Obs.	CD F	Hansen J
Panel A. Gross fixed capital formation						
OF	0.163***	(0.038)	136	1548	36.51	0.00
OOF	0.391***	(0.099)	136	1548	38.12	0.90
ODA	0.207***	(0.054)	136	1548	29.05	0.31
Panel B. Gross fixed private capital formation						
OF	0.226*	(0.122)	106	1251	30.62	0.63
OOF	0.580*	(0.339)	106	1251	16.20	0.94
ODA	0.319*	(0.171)	106	1251	29.81	0.49
Panel C. Foreign direct investment inflows						
OF	0.424***	(0.110)	153	1945	53.82	0.01
OOF	0.671***	(0.227)	153	1945	55.81	0.00
ODA	0.584***	(0.159)	153	1945	43.83	0.50

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Table 5: Chinese Financing and Components of GDP

Table: (Continued from previous slide)

	Coeff.	SE	Countries	Obs.	CD F	Hansen J
Panel D. Imports						
OF	0.138***	(0.033)	144	1590	38.84	0.00
OOF	0.372***	(0.104)	144	1590	39.66	0.41
ODA	0.156***	(0.044)	144	1590	30.30	0.02
Panel E. Exports						
OF	0.088**	(0.035)	144	1590	38.84	0.00
OOF	0.205**	(0.091)	144	1590	39.66	0.99
ODA	0.120**	(0.051)	144	1590	30.30	0.76
Panel F. Consumption, overall						
OF	0.084***	(0.018)	135	1537	38.01	0.00
OOF	0.224***	(0.048)	135	1537	39.40	0.43
ODA	0.099***	(0.025)	135	1537	29.81	0.05

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Table 5: Chinese Financing and Components of GDP

Table: (Continued from previous slide)

	Coeff.	SE	Countries	Obs.	CD F	Hansen J
Panel G. Consumption, household						
OF	0.088***	(0.020)	136	1538	37.95	0.00
OOF	0.232***	(0.052)	136	1538	39.37	0.53
ODA	0.098***	(0.026)	136	1538	29.77	0.01
Panel H. Consumption, government						
OF	0.073***	(0.021)	136	1538	37.95	0.00
OOF	0.221***	(0.069)	136	1538	39.37	0.14
ODA	0.112***	(0.033)	136	1538	29.77	0.28
Panel I. Savings						
OF	0.161***	(0.051)	133	1486	39.43	0.00
OOF	0.446***	(0.110)	133	1486	27.18	0.27
ODA	0.199**	(0.081)	133	1486	34.26	0.04

Table 5: Chinese Financing and Components of GDP

TABLE 5—CHINESE OFFICIAL FINANCING AND COMPONENTS OF GDP (2SLS, 2002–2016)

	Coefficient	Standard error	Countries	Observations	Cragg-Donald <i>F</i> -statistic	Kleibergen-Paap <i>F</i> -statistic	Hansen <i>J</i> -statistic
<i>Panel A. Gross fixed capital formation</i>							
OF	0.024	(0.011)	117	1,481	30.00	33.55	0.22
OOFF	−0.014	(0.027)	117	1,481	29.19	22.84	0.44
ODA	0.04	(0.015)	117	1,481	28.84	29.55	0.07
<i>Panel B. Gross fixed private capital formation</i>							
OF	0.083	(0.071)	103	1,215	21.46	34.79	0.37
OOFF	0.173	(0.210)	103	1,215	14.86	22.52	0.87
ODA	0.112	(0.096)	103	1,215	26.00	28.71	0.31
<i>Panel C. Foreign direct investment inflows</i>							
OF	0.030	(0.050)	149	1,852	36.79	32.00	0.72
OOFF	0.089	(0.073)	149	1,852	47.14	14.43	0.95
ODA	0.038	(0.061)	149	1,852	34.83	31.97	0.65
<i>Panel D. Imports</i>							
OF	0.012	(0.009)	121	1,532	32.57	32.10	0.86
OOFF	−0.016	(0.023)	121	1,532	32.53	24.72	0.00
ODA	0.024	(0.012)	121	1,532	30.11	27.53	0.18
<i>Panel E. Exports</i>							
OF	0.003	(0.010)	121	1,532	32.57	32.10	0.43
OOFF	−0.038	(0.022)	121	1,532	32.53	24.72	0.36
ODA	0.016	(0.013)	121	1,532	30.11	27.53	0.72
<i>Panel F. Consumption, overall</i>							
OF	0.005	(0.003)	116	1,470	31.67	36.03	0.01
OOFF	0.004	(0.009)	116	1,470	31.67	25.13	0.93
ODA	0.006	(0.004)	116	1,470	29.38	30.30	0.00
<i>Panel G. Consumption, household</i>							
OF	0.004	(0.003)	116	1,470	31.67	36.03	0.02
OOFF	0.007	(0.009)	116	1,470	31.67	25.13	0.49
ODA	0.006	(0.005)	116	1,470	29.38	30.30	0.01
<i>Panel H. Consumption, government</i>							
OF	0.005	(0.009)	116	1,470	31.67	36.03	0.10
OOFF	0.007	(0.029)	116	1,470	31.67	25.13	0.90
ODA	0.007	(0.013)	116	1,470	29.38	30.30	0.14
<i>Panel I. Savings</i>							
OF	−0.032	(0.020)	127	1,404	26.72	30.51	0.86
OOFF	−0.005	(0.035)	127	1,404	20.90	26.02	0.56
ODA	−0.041	(0.032)	127	1,404	27.49	24.10	0.94

Notes: Each row represents one regression. The dependent variable is—in first differences, logged constant 2010 US dollars, with a value of one added before taking logs—indicated in the panel header. The variable of interest, *Chinese OF* ($t - 2$), denotes Chinese development finance commitments lagged by two years and is measured as a project count. In separate rows, we show results for all official finance (*OF*) projects, Other Official Flows (*OOFF*) projects, and Official Development Assistance (*ODA*) projects. All regressions include $(\log) Population (t - 1)$ and country- and year-fixed effects as control variables. Standard errors are in parentheses.

Our Data Extension (Addressing IV Limitations)

- **The Data Gap:** While constructing the instrumental variables, we identified a hard limitation in the author-provided dataset: the `input_factor` variable (measuring China's material overproduction) ended abruptly in 2013.
- **Our Synthesized Solution:** To maximize the utility of the dataset, we synthesized the missing years:
 - We created an extended variable (`iv_input_of_ext`) by forward-filling the 2013 values through 2014, 2015, and 2016.
- **Impact of the Extension:** This allowed us to successfully:
 - Reconstruct the interaction instruments (`iv_input_of`, `iv_input_oda`, `iv_input_oof`) up to 2016.
 - Execute lead/lag regressions without dropping the most recent, highly relevant project observations due to missing data.

Appendix Figure B1: Project Status by Year

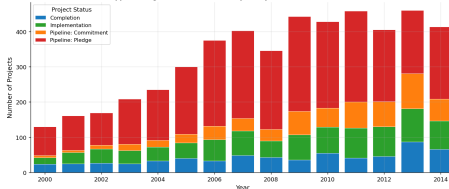
- This figure tracks the lifecycle of Chinese finance, showing a clear upward trend in the number of projects moving from “commitment” to “implementation” and “completion” over the 2000–2014 period.

Appendix Figure B3: Sectoral Distribution

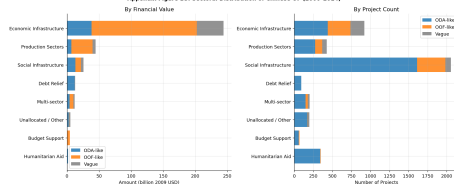
- The sectoral breakdown confirms that Chinese development finance is heavily concentrated in **infrastructure**—specifically transportation, energy, and communications—rather than traditional social sectors.

Appendix Figures B1 & B3: Project Status and Sectoral Distribution

Appendix Figure B1: Chinese OF Projects by Status and Year (2000-2014)



Appendix Figure B3: Sectoral Distribution of Chinese OF (2000-2014)

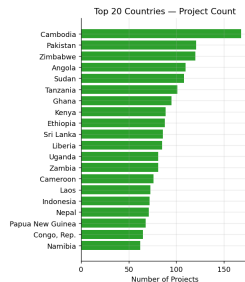
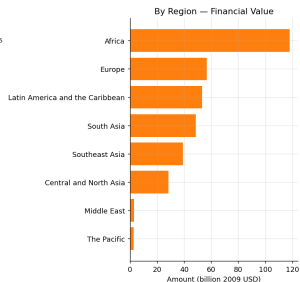
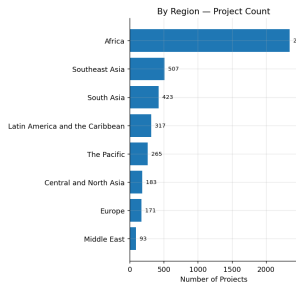


Appendix Figures B5 & B6: Regional and Country Distribution

- **Regional Takeaway:** Africa dominates in terms of absolute project count, but the financial volume (measured in billions of USD) is more evenly distributed across Asia and Africa due to massive individual infrastructure loans.
- **Country-Level Takeaway:** The Top 20 recipient breakdown reveals that countries like **Russia, Pakistan, and Angola** are the primary destinations for the highest financial volumes.

Appendix Figures B5 & B6: Geographic Distribution

Appendix Figures B5-B6: Geographic Distribution of Chinese OF (2000-2014)



Thank You!